Madhav Anand Menon

+1 (646) 938-7079 | madhavanandmenon@gmail.com | linkedin.com/in/madhavmenon | github.com/MadhavMenon10

Education

University of Illinois Urbana-Champaign Expected May 2028
B.S. Computer Science and Physics; Minor in Mathematics (James Scholar) **GPA: 3.97/4.0**

- Yee Seung Ng Award; Harold and Ruth Hayward Scholarship; Dean's List; Tau Beta Pi Honours Society
- Relevant Coursework: Object Oriented Programming, Data Structures and Algorithms, Computer Architecture, Linear Algebra (Upcoming: Operating Systems, Numerical Methods, Databases)

Experience

Disruption Lab Feb. 2026 - Present
Software Engineer Champaign, IL, USA

- Built a 5-stage Canvas-to-PostgreSQL RAG pipeline supporting 3 file formats (PDF/DOCX/TXT) with 1024-dim pgvector embeddings and sliding-window chunking via AWS Bedrock, reducing retrieval latency by 35%
- Designed a prompt-engineered skill extractor on Claude Haiku (AWS Bedrock) with NACE framework alignment, returning deduplicated skill taxonomies across 100+ courses per API call — reducing manual tagging effort by 98%
- Shipped a 9-endpoint FastAPI backend with stateless HMAC-SHA256 authentication (8-hr TTL, zero external dependencies) and dual-mode search (full-text PostgreSQL and pgvector cosine similarity) supporting 120+ concurrent users

Algoverse AI Sep. 2025 - May 2026
LLM Researcher San Francisco, CA, USA (Remote)

- Built DeployStega, an LLM steganographic communication framework using token-binning to encode up to 8 bits/word, achieving 90%+ word inclusion fidelity validated against a 13M-event corpus spanning 695K+ real GitHub users
- Merived statistical priors from GH Archive (179s median inter-event gap; 0.899 commit-to-commit transition probability; 78.5% commit share) to constrain synthetic trace generation within an empirically normal behavioral envelope, ensuring population-level indistinguishability
- Evaluated detection resistance against 5 adversarial classifiers (LR, RF, SVM, TS-CNN, BERT-base) on 4,000 synthetic users; adapted a differential-privacy ε metric from TPR/FPR to quantify population-level indistinguishability, achieving $\varepsilon < 1.173$

DigiAlert Jun. 2025 - Aug. 2025
Software Engineer Intern Santa Clara, CA, USA (Remote)

- Built a Flask-based AWS CSPM dashboard aggregating security metrics across 4 AWS services with Chart.js visualizations, cutting CSPM tooling costs by 45%
- Built a concurrent data pipeline (boto3, threading) processing 500+ CloudTrail events per refresh with real-time CIS v5.0.0 risk classification, reducing pipeline execution time by 35% vs. sequential baseline]
- Deployed dashboard via nginx/gunicorn with REST API integration and TTL-based cache invalidation, reducing stale data exposure and cutting redundant API calls by 27%

Projects

GPU-Accelerated Ray Tracer | *CUDA, C++*

- Engineered a real-time ray tracer in CUDA achieving a 40x speedup over a single-threaded CPU baseline, rendering scenes at 60+ FPS at 1080p on an NVIDIA RTX 3060
- Integrated a BVH acceleration structure that reduced ray-intersection tests by 80%, cutting average frame render time from 95ms to 14ms
- Optimized memory coalescing and warp utilization to raise kernel occupancy from 50% to 88%, enabling rendering of scenes with 1M+ triangles and 10M+ rays per frame

Skills

Languages: C++; C; Python; Java; SQL; Verilog; MIPS Assembly
Libraries and Frameworks: CUDA; PyTorch; FastAPI; Flask; pandas; scikit-learn; NumPy; Matplotlib
Tools: NVIDIA Nsight; AWS; Azure; GCP; Git; TeX/LaTeX; Docker; Valgrind, Mathematica, MATLAB
Other Activities: CS 124 Project Manager; TEDx Speaker; ACM Corporate Team Member